

Supplementary Materials for "Provable Lipschitz Continuity Network for Universal CT Reconstruction"

This document is organized as follows. We first provide the detailed proofs in Section I, and then report more visualization results in Section II. Section III presents generalization to extremely low-view scenarios. Section IV provides inference time, network parameters and FLOPs.

I. THE PROOFS

A. Proof of Lemma 1

According to the definition of a single-layer tight frame network, for $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we have

$$\begin{aligned} & \| \mathbf{W}^T \mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathbf{W}^T \mathcal{S}_A(\mathbf{W}\mathbf{x}) \|_2^2 \\ &= \| \mathbf{W}^T [\mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x})] \|_2^2 \\ &\leq \| \mathbf{W}^T \|_2^2 \| \mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x}) \|_2^2 \\ &= \lambda_{\max}(\mathbf{W}\mathbf{W}^T) \| \mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x}) \|_2^2 \end{aligned} \quad (1)$$

where $\lambda_{\max}(\cdot)$ denotes the maximum eigenvalue.

Let $(\mathbf{W}\mathbf{x})_i$ represents the i -th element of $\mathbf{W}\mathbf{x}$. The soft thresholding operator $\mathcal{S}_A(\mathbf{W}\mathbf{x})$ is defined as

$$\mathcal{S}_A(\mathbf{W}\mathbf{x}) = \begin{cases} (\mathbf{W}\mathbf{x})_i + \lambda_i, & (\mathbf{W}\mathbf{x})_i < -\lambda_i \\ 0, & |(\mathbf{W}\mathbf{x})_i| \leq \lambda_i \\ (\mathbf{W}\mathbf{x})_i - \lambda_i, & (\mathbf{W}\mathbf{x})_i > \lambda_i. \end{cases} \quad (2)$$

Let $(\mathbf{W}\mathbf{y})_i$ represents the i -th element of $\mathbf{W}\mathbf{y}$. The soft thresholding operator $\mathcal{S}_A(\mathbf{W}\mathbf{y})$ is defined as

$$\mathcal{S}_A(\mathbf{W}\mathbf{y}) = \begin{cases} (\mathbf{W}\mathbf{y})_i + \lambda_i, & (\mathbf{W}\mathbf{y})_i < -\lambda_i \\ 0, & |(\mathbf{W}\mathbf{y})_i| \leq \lambda_i \\ (\mathbf{W}\mathbf{y})_i - \lambda_i, & (\mathbf{W}\mathbf{y})_i > \lambda_i. \end{cases} \quad (3)$$

Consider the following scenarios:

C1: $(\mathbf{W}\mathbf{x})_i < (\mathbf{W}\mathbf{y})_i$

- If $(\mathbf{W}\mathbf{x})_i < (\mathbf{W}\mathbf{y})_i < -\lambda_i$, then $\mathcal{S}_A(\mathbf{W}\mathbf{x}) = \mathbf{W}\mathbf{x} + \mathbf{A}$, $\mathcal{S}_A(\mathbf{W}\mathbf{y}) = \mathbf{W}\mathbf{y} + \mathbf{A}$.

$$\begin{aligned} & \| \mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x}) \|_2^2 \\ &= \| (\mathbf{W}\mathbf{y} + \mathbf{A}) - (\mathbf{W}\mathbf{x} + \mathbf{A}) \|_2^2 \\ &= \| \mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x} \|_2^2 \\ &\leq \| \mathbf{W} \|_2^2 \| \mathbf{y} - \mathbf{x} \|_2^2 \\ &= \| \mathbf{y} - \mathbf{x} \|_2^2. \end{aligned} \quad (4)$$

- If $(\mathbf{W}\mathbf{x})_i < -\lambda_i < (\mathbf{W}\mathbf{y})_i < \lambda_i$, then $\mathcal{S}_A(\mathbf{W}\mathbf{x}) = \mathbf{W}\mathbf{x} + \mathbf{A}$, $\mathcal{S}_A(\mathbf{W}\mathbf{y}) = \mathbf{0}$.

$$\begin{aligned} & \| \mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x}) \|_2^2 \\ &= \| \mathbf{0} - (\mathbf{W}\mathbf{x} + \mathbf{A}) \|_2^2 \\ &= \| -\mathbf{W}\mathbf{x} - \mathbf{A} \|_2^2 \\ &\leq \| \mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x} \|_2^2 \\ &\leq \| \mathbf{W} \|_2^2 \| \mathbf{y} - \mathbf{x} \|_2^2 \\ &= \| \mathbf{y} - \mathbf{x} \|_2^2. \end{aligned} \quad (5)$$

- If $(W\mathbf{x})_i < -\lambda_i < \lambda_i < (W\mathbf{y})_i$, then $\mathcal{S}_\Lambda(W\mathbf{x}) = W\mathbf{x} + \Lambda$, $\mathcal{S}_\Lambda(W\mathbf{y}) = W\mathbf{y} - \Lambda$.

$$\begin{aligned}
& \|S_A(\mathbf{W}\mathbf{y}) - S_A(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} - \mathbf{A}) - (\mathbf{W}\mathbf{x} + \mathbf{A})\|_2^2 \\
&= \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x} - 2\mathbf{A}\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{6}$$

- If $-\lambda_i \leq (Wx)_i < (Wy)_i \leq \lambda_i$, then $S_A(Wx) = 0$, $S_A(Wy) = 0$.

$$\begin{aligned} & \|\mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x})\|_2^2 \\ &= \|\mathbf{0} - \mathbf{0}\|_2^2 \\ &= 0. \end{aligned} \tag{7}$$

- If $-\lambda_i \leq (\mathbf{W}\mathbf{x})_i \leq \lambda_i < (\mathbf{W}\mathbf{y})_i$, then $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{x}) = 0$, $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) = \mathbf{W}\mathbf{y} - \Lambda$.

$$\begin{aligned}
& \|\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} - \Lambda) - \mathbf{0}\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{8}$$

- If $\lambda_i < (\mathbf{W}\mathbf{x})_i < (\mathbf{W}\mathbf{y})_i$, then $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{x}) = \mathbf{W}\mathbf{x} - \Lambda$, $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) = \mathbf{W}\mathbf{y} - \Lambda$.

$$\begin{aligned}
& \|\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} - \Lambda) - (\mathbf{W}\mathbf{x} - \Lambda)\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{9}$$

C2: $(Wy)_i < (Wx)_i$

- If $(W\mathbf{y})_i < (W\mathbf{x})_i < -\lambda_i$, then $\mathcal{S}_\Lambda(W\mathbf{x}) = W\mathbf{x} + \Lambda$, $\mathcal{S}_\Lambda(W\mathbf{y}) = W\mathbf{y} + \Lambda$.

$$\begin{aligned}
& \|\mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} + \mathbf{A}) - (\mathbf{W}\mathbf{x} + \mathbf{A})\|_2^2 \\
&= \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{10}$$

- If $(W\mathbf{y})_i < -\lambda_i < (W\mathbf{x})_i < \lambda_i$, then $S_A(W\mathbf{x}) = 0$, $S_A(W\mathbf{y}) = W\mathbf{y} + \Lambda$.

$$\begin{aligned}
& \|\mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} + \mathbf{A}) - \mathbf{0}\|_2^2 \\
&= \|\mathbf{W}\mathbf{y} + \mathbf{A}\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{11}$$

- If $(W\mathbf{y})_i < -\lambda_i < \lambda_i < (W\mathbf{x})_i$, then $\mathcal{S}_A(W\mathbf{x}) = W\mathbf{x} - \Lambda$, $\mathcal{S}_A(W\mathbf{y}) = W\mathbf{y} + \Lambda$.

$$\begin{aligned} & \|\mathcal{S}_A(\mathbf{W}\mathbf{y}) - \mathcal{S}_A(\mathbf{W}\mathbf{x})\|_2^2 \\ &= \|(\mathbf{W}\mathbf{y} + \mathbf{\Lambda}) - (\mathbf{W}\mathbf{x} - \mathbf{\Lambda})\|_2^2 \\ &= \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x} + 2\mathbf{\Lambda}\|_2^2 \end{aligned}$$

$$\begin{aligned}
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{12}$$

- If $-\lambda_i \leq (\mathbf{W}\mathbf{y})_i < (\mathbf{W}\mathbf{x})_i \leq \lambda_i$, then $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{x}) = \mathbf{0}$, $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) = \mathbf{0}$.

$$\begin{aligned}
&\|\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|\mathbf{0} - \mathbf{0}\|_2^2 \\
&= 0.
\end{aligned} \tag{13}$$

- If $-\lambda_i \leq (\mathbf{W}\mathbf{y})_i \leq \lambda_i < (\mathbf{W}\mathbf{x})_i$, then $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{x}) = \mathbf{W}\mathbf{x} - \Lambda$, $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) = \mathbf{0}$.

$$\begin{aligned}
&\|\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|\mathbf{0} - (\mathbf{W}\mathbf{x} - \Lambda)\|_2^2 \\
&= \|\mathbf{W}\mathbf{x} - \Lambda\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{x} - \mathbf{W}\mathbf{y}\|_2^2 \\
&\leq \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{14}$$

- If $\lambda_i < (\mathbf{W}\mathbf{y})_i < (\mathbf{W}\mathbf{x})_i$, then $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{x}) = \mathbf{W}\mathbf{x} - \Lambda$, $\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) = \mathbf{W}\mathbf{y} - \Lambda$.

$$\begin{aligned}
&\|\mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \\
&= \|(\mathbf{W}\mathbf{y} - \Lambda) - (\mathbf{W}\mathbf{x} - \Lambda)\|_2^2 \\
&= \|\mathbf{W}\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 \\
&\leq \|\mathbf{W}\|_2^2 \|\mathbf{y} - \mathbf{x}\|_2^2 \\
&= \|\mathbf{y} - \mathbf{x}\|_2^2.
\end{aligned} \tag{15}$$

From the above inequalities, it can be concluded that

$$\|\mathbf{W}^T \mathcal{S}_\Lambda(\mathbf{W}\mathbf{y}) - \mathbf{W}^T \mathcal{S}_\Lambda(\mathbf{W}\mathbf{x})\|_2^2 \leq \lambda_{\max}(\mathbf{W}\mathbf{W}^T) \|\mathbf{y} - \mathbf{x}\|_2^2.$$

Therefore, the single tight frame N is ε -Lipschitzian, where $\varepsilon = \sqrt{\lambda_{\max}(\mathbf{W}\mathbf{W}^T)}$. ■

B. Proof of Theorem1

The structure of Proms-Net can be expressed as

$$M(\mathbf{x}) = \alpha \odot N_1(\mathbf{x}) + \beta \odot N_2(\mathbf{x}) + \gamma \odot N_3(\mathbf{x}), \tag{16}$$

where $\gamma = \mathbf{1} - \alpha - \beta$, $\odot$ denotes element-wise multiplication.

Then, for $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we can bound the difference between $M(\mathbf{y})$ and $M(\mathbf{x})$ as follows

$$\begin{aligned}
&\|M(\mathbf{y}) - M(\mathbf{x})\|_2 \\
&= \|\alpha_{\mathbf{y}} \odot N_1(\mathbf{y}) + \beta_{\mathbf{y}} \odot N_2(\mathbf{y}) + \gamma_{\mathbf{y}} \odot N_3(\mathbf{y}) - \alpha_{\mathbf{x}} \odot N_1(\mathbf{x}) - \beta_{\mathbf{x}} \odot N_2(\mathbf{x}) - \gamma_{\mathbf{x}} \odot N_3(\mathbf{x})\|_2 \\
&= \|\alpha_{\mathbf{y}} \odot [N_1(\mathbf{y}) - N_1(\mathbf{x})] + \beta_{\mathbf{y}} \odot [N_2(\mathbf{y}) - N_2(\mathbf{x})] + \gamma_{\mathbf{y}} \odot [N_3(\mathbf{y}) - N_3(\mathbf{x})] \\
&\quad + (\alpha_{\mathbf{y}} - \alpha_{\mathbf{x}}) \odot N_1(\mathbf{x}) + (\beta_{\mathbf{y}} - \beta_{\mathbf{x}}) \odot N_2(\mathbf{x}) + (\gamma_{\mathbf{y}} - \gamma_{\mathbf{x}}) \odot N_3(\mathbf{x})\|_2.
\end{aligned} \tag{17}$$

where $\alpha_{\mathbf{x}}$, $\alpha_{\mathbf{y}}$, $\beta_{\mathbf{x}}$, $\beta_{\mathbf{y}}$, $\gamma_{\mathbf{x}}$ and $\gamma_{\mathbf{y}}$ are weighting coefficients.

Next, applying the triangle inequality, we obtain

$$\begin{aligned}
&\|M(\mathbf{y}) - M(\mathbf{x})\|_2 \\
&\leq \|\alpha_{\mathbf{y}} \odot [N_1(\mathbf{y}) - N_1(\mathbf{x})]\|_2 + \|\beta_{\mathbf{y}} \odot [N_2(\mathbf{y}) - N_2(\mathbf{x})]\|_2 + \|\gamma_{\mathbf{y}} \odot [N_3(\mathbf{y}) - N_3(\mathbf{x})]\|_2 \\
&\quad + \|(\alpha_{\mathbf{y}} - \alpha_{\mathbf{x}}) \odot N_1(\mathbf{x})\|_2 + \|(\beta_{\mathbf{y}} - \beta_{\mathbf{x}}) \odot N_2(\mathbf{x})\|_2 + \|(\gamma_{\mathbf{y}} - \gamma_{\mathbf{x}}) \odot N_3(\mathbf{x})\|_2.
\end{aligned} \tag{18}$$

Using the inequality $\|A \odot B\|_2 \leq \|A\|_\infty \|B\|_2$ for the element-wise product, we have

$$\begin{aligned} & \|M(\mathbf{y}) - M(\mathbf{x})\|_2 \\ & \leq \|\alpha_{\mathbf{y}}\|_\infty \|N_1(\mathbf{y}) - N_1(\mathbf{x})\|_2 + \|\beta_{\mathbf{y}}\|_\infty \|N_2(\mathbf{y}) - N_2(\mathbf{x})\|_2 + \|\gamma_{\mathbf{y}}\|_\infty \|N_3(\mathbf{y}) - N_3(\mathbf{x})\|_2 \\ & \quad + \|\alpha_{\mathbf{y}} - \alpha_{\mathbf{x}}\|_\infty \|N_1(\mathbf{x})\|_2 + \|\beta_{\mathbf{y}} - \beta_{\mathbf{x}}\|_\infty \|N_2(\mathbf{x})\|_2 + \|\gamma_{\mathbf{y}} - \gamma_{\mathbf{x}}\|_\infty \|N_3(\mathbf{x})\|_2. \end{aligned} \quad (19)$$

Since each of the individual components $N_1(\cdot)$, $N_2(\cdot)$, and $N_3(\cdot)$ is a single tight frame network, we deduce from **Lemma1** that $N_1(\cdot)$, $N_2(\cdot)$, and $N_3(\cdot)$ are Lipschitz continuous with constants ε_1 , ε_2 , and ε_3 , respectively. Specifically, $N_1(\cdot)$ is ε_1 -Lipschitzian, where $\varepsilon_1 = \sqrt{\lambda_{\max}(\mathbf{W}_1 \mathbf{W}_1^\top)}$. $N_2(\cdot)$ is ε_2 -Lipschitzian, where $\varepsilon_2 = \sqrt{\lambda_{\max}(\mathbf{W}_2 \mathbf{W}_2^\top)}$. $N_3(\cdot)$ is ε_3 -Lipschitzian, where $\varepsilon_3 = \sqrt{\lambda_{\max}(\mathbf{W}_3 \mathbf{W}_3^\top)}$.

Considering $\alpha_{\mathbf{x}}$, $\alpha_{\mathbf{y}}$, $\beta_{\mathbf{x}}$, $\beta_{\mathbf{y}}$, $\gamma_{\mathbf{x}}$ and $\gamma_{\mathbf{y}}$ are spatial-varying weighting coefficients generated by the Weight Module with the constraint that all elements lie in $[0, 1]$, we derive

$$\|\alpha_{\mathbf{y}} - \alpha_{\mathbf{x}}\|_\infty \leq 1, \quad \|\beta_{\mathbf{y}} - \beta_{\mathbf{x}}\|_\infty \leq 1, \quad \|\gamma_{\mathbf{y}} - \gamma_{\mathbf{x}}\|_\infty \leq 1. \quad (20)$$

Thus, the difference between $M(\mathbf{y})$ and $M(\mathbf{x})$ is bounded by

$$\begin{aligned} & \|M(\mathbf{y}) - M(\mathbf{x})\|_2 \\ & \leq \varepsilon_1 \|\mathbf{y} - \mathbf{x}\|_2 + \varepsilon_2 \|\mathbf{y} - \mathbf{x}\|_2 + \varepsilon_3 \|\mathbf{y} - \mathbf{x}\|_2 \\ & \quad + \|N_1(\mathbf{x})\|_2 + \|N_2(\mathbf{x})\|_2 + \|N_3(\mathbf{x})\|_2. \end{aligned} \quad (21)$$

Now, we use the fact that Lipschitz continuous functions have bounded growth. Specifically, for any Lipschitz continuous function f with constant L_f and any point $\mathbf{x}_0$, we have

$$\|f(\mathbf{x})\|_2 \leq \|f(\mathbf{x}_0)\|_2 + L_f \|\mathbf{x} - \mathbf{x}_0\|_2. \quad (22)$$

Choosing $\mathbf{x}_0 = \mathbf{0}$ and applying this to each N_i , we obtain

$$\|N_i(\mathbf{x})\|_2 \leq \|N_i(\mathbf{0})\|_2 + \varepsilon_i \|\mathbf{x}\|_2. \quad (23)$$

According to the definition of the sparsifying frame network, $N_i(\mathbf{0}) = \mathbf{0}$, and thus $\|N_i(\mathbf{0})\|_2 = 0$. Then

$$\|N_i(\mathbf{x})\|_2 \leq \varepsilon_i \|\mathbf{x}\|_2. \quad (24)$$

Substituting back into inequality (21), we have

$$\begin{aligned} & \|M(\mathbf{y}) - M(\mathbf{x})\|_2 \\ & \leq \varepsilon_1 \|\mathbf{y} - \mathbf{x}\|_2 + \varepsilon_2 \|\mathbf{y} - \mathbf{x}\|_2 + \varepsilon_3 \|\mathbf{y} - \mathbf{x}\|_2 \\ & \quad + \varepsilon_1 \|\mathbf{x}\|_2 + \varepsilon_2 \|\mathbf{x}\|_2 + \varepsilon_3 \|\mathbf{x}\|_2 \\ & = (\varepsilon_1 + \varepsilon_2 + \varepsilon_3) (\|\mathbf{y} - \mathbf{x}\|_2 + \|\mathbf{x}\|_2). \end{aligned} \quad (25)$$

In practical CT reconstructions, the input image $\mathbf{x}$ is normalized to a finite range between 0 and 1, and the total number of pixels is fixed. Since the $\|\mathbf{x}\|_2$ is the square root of the sum of the squared pixel values, the sum of the squares of a finite number of bounded values must be finite. Therefore, the norm is bounded. Let R be an upper bound on $\|\mathbf{x}\|_2$. Then

$$\|M(\mathbf{y}) - M(\mathbf{x})\|_2 \leq (\varepsilon_1 + \varepsilon_2 + \varepsilon_3) (\|\mathbf{y} - \mathbf{x}\|_2 + R). \quad (26)$$

To obtain a Lipschitz constant, we note that for any $\mathbf{x}, \mathbf{y}$ with $\|\mathbf{x}\|_2 \leq R$ and $\|\mathbf{y}\|_2 \leq R$, we derive

$$\|M(\mathbf{y}) - M(\mathbf{x})\|_2 \leq \left[(\varepsilon_1 + \varepsilon_2 + \varepsilon_3) \left(1 + \frac{R}{\|\mathbf{y} - \mathbf{x}\|_2} \right) \right] \|\mathbf{y} - \mathbf{x}\|_2. \quad (27)$$

When $\mathbf{y} = \mathbf{x}$, both sides of the Lipschitz inequality are 0, which clearly holds. For the case $\mathbf{y} \neq \mathbf{x}$, in practical CT reconstructions, there exists a minimum distance $\delta > 0$ nature of digital images and physical constraints. Therefore, we have $\|\mathbf{y} - \mathbf{x}\|_2 \geq \delta$, which implies

$$\|M(\mathbf{y}) - M(\mathbf{x})\|_2 \leq \left[(\varepsilon_1 + \varepsilon_2 + \varepsilon_3) \left(1 + \frac{R}{\delta} \right) \right] \|\mathbf{y} - \mathbf{x}\|_2. \quad (28)$$

Therefore, the proposed Proms-Net $M(\cdot)$ is Lipschitz continuous with constant

$$\varepsilon' = (\varepsilon_1 + \varepsilon_2 + \varepsilon_3) \left(1 + \frac{R}{\delta} \right). \quad (29)$$

■

C. Proof of Lemma2

Let $g(t) = F(t(\mathbf{x} - \mathbf{y}) + \mathbf{y})$, we have

$$F(\mathbf{x}) = g(1), F(\mathbf{y}) = g(0), \quad (30)$$

$$g'(t) = \nabla F(t(\mathbf{x} - \mathbf{y}) + \mathbf{y})^T (\mathbf{x} - \mathbf{y}). \quad (31)$$

Hence, we can derive that

$$\begin{aligned} & F(\mathbf{x}) - F(\mathbf{y}) - \langle \nabla F(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \\ &= g(1) - g(0) - \nabla F(\mathbf{y})^T (\mathbf{x} - \mathbf{y}) \\ &= \int_0^1 g'(t) dt - \nabla F(\mathbf{y})^T (\mathbf{x} - \mathbf{y}) \\ &= \int_0^1 [\nabla F(t(\mathbf{x} - \mathbf{y}) + \mathbf{y}) - \nabla F(\mathbf{y})]^T (\mathbf{x} - \mathbf{y}) dt \\ &\leq \int_0^1 \|\nabla F(t(\mathbf{x} - \mathbf{y}) + \mathbf{y}) - \nabla F(\mathbf{y})\|_2 \cdot \|\mathbf{x} - \mathbf{y}\|_2 dt \\ &\leq \int_0^1 L \|t(\mathbf{x} - \mathbf{y})\|_2 \cdot \|\mathbf{x} - \mathbf{y}\|_2 dt \\ &= \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|_2^2. \end{aligned}$$

Therefore, we obtain

$$F(\mathbf{x}) \leq F(\mathbf{y}) + \langle \nabla F(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|_2^2. \quad (32)$$

■

D. Proof of Lemma3

In our optimization problem, we set a convex smooth ℓ_2 data-fidelity term $f(\mathbf{x}) = \frac{1}{2} \|\mathbf{y} - \mathbf{P}\mathbf{x}\|_2^2$. A smooth function refers to a function that is infinitely differentiable. Thus, there exists a constant $L_f > 0$ such that the gradient ∇f of $f(\cdot)$ satisfies

$$\|\nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_2)\|_2 \leq L_f \|\mathbf{x}_1 - \mathbf{x}_2\|_2, \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n.$$

The gradient of our objective function, i.e., ∇F , can be calculated as follows

$$\nabla F(\mathbf{x}) = -\mathbf{P}^T(\mathbf{y} - \mathbf{P}\mathbf{x}) + \lambda M(\mathbf{x}), \quad (33)$$

where $M(\cdot)$ is ε' -Lipschitzian based on **Lemma 1**. Then, for $\forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$

$$\begin{aligned} & \|\nabla F(\mathbf{x}_1) - \nabla F(\mathbf{x}_2)\|_2 \\ &= \|-\mathbf{P}^T(\mathbf{y} - \mathbf{P}\mathbf{x}_1) + \lambda M(\mathbf{x}_1) - (-\mathbf{P}^T(\mathbf{y} - \mathbf{P}\mathbf{x}_2) + \lambda M(\mathbf{x}_2))\|_2 \\ &= \|\mathbf{P}^T \mathbf{P}(\mathbf{x}_1 - \mathbf{x}_2) + \lambda (M(\mathbf{x}_1) - M(\mathbf{x}_2))\|_2 \\ &\leq \|\mathbf{P}^T \mathbf{P}(\mathbf{x}_1 - \mathbf{x}_2)\|_2 + \lambda \|M(\mathbf{x}_1) - M(\mathbf{x}_2)\|_2 \\ &\leq \|\mathbf{P}^T \mathbf{P}\|_2 \cdot \|\mathbf{x}_1 - \mathbf{x}_2\|_2 + \varepsilon' \lambda \|\mathbf{x}_1 - \mathbf{x}_2\|_2 \\ &\leq (L' + \varepsilon' \lambda) \|\mathbf{x}_1 - \mathbf{x}_2\|_2, \end{aligned} \quad (34)$$

where $L' = \lambda_{\max}(\mathbf{P}^T \mathbf{P})$. Therefore, ∇F is Lipschitz continuous with a constant $L = \lambda_{\max}(\mathbf{P}^T \mathbf{P}) + \varepsilon' \lambda$. ■

E. Proof of Theorem2

Let $\mathbf{y} = \mathbf{x}^{(k)}, \mathbf{x} = \mathbf{x}^{(k+1)}$ in inequality (32), it holds that for any $0 < \eta \leq \frac{1}{L}$, we have

$$\begin{aligned} F(\mathbf{x}^{(k+1)}) &\leq F(\mathbf{x}^{(k)}) + \langle \nabla F(\mathbf{x}^{(k)}), \mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} \rangle + \frac{L}{2} \|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2^2 \\ &= F(\mathbf{x}^{(k)}) + \langle \nabla F(\mathbf{x}^{(k)}), -\eta \nabla F(\mathbf{x}^{(k)}) \rangle + \frac{L}{2} \|- \eta \nabla F(\mathbf{x}^{(k)})\|_2^2 \\ &= F(\mathbf{x}^{(k)}) - \eta \|\nabla F(\mathbf{x}^{(k)})\|_2^2 + \eta^2 \frac{L}{2} \|\nabla F(\mathbf{x}^{(k)})\|_2^2 \end{aligned}$$

$$\begin{aligned}
&= F(\mathbf{x}^{(k)}) - \eta \left(1 - \frac{\eta L}{2}\right) \|\nabla F(\mathbf{x}^{(k)})\|_2^2 \\
&\leq F(\mathbf{x}^{(k)}) - \frac{\eta}{2} \|\nabla F(\mathbf{x}^{(k)})\|_2^2.
\end{aligned} \tag{35}$$

Therefore, for any $k \geq 0$, we obtain

$$\|\nabla F(\mathbf{x}^{(k)})\|_2^2 \leq \frac{2}{\eta} \left(F(\mathbf{x}^{(k)}) - F(\mathbf{x}^{(k+1)})\right). \tag{36}$$

By summing the above inequality for $i = 0, \dots, k-1$, we derive

$$\begin{aligned}
\sum_{i=0}^{k-1} \|\nabla F(\mathbf{x}^{(i)})\|_2^2 &\leq \frac{2}{\eta} \sum_{i=0}^{k-1} \left(F(\mathbf{x}^{(i)}) - F(\mathbf{x}^{(i+1)})\right) \\
&= \frac{2}{\eta} \left(F(\mathbf{x}^{(0)}) - F(\mathbf{x}^{(k)})\right).
\end{aligned} \tag{37}$$

As $F^* \leq F(\mathbf{x}^{(k)})$ holds for any $k \geq 0$, it can be concluded that

$$\sum_{i=0}^{k-1} \|\nabla F(\mathbf{x}^{(i)})\|_2^2 \leq \frac{2}{\eta} \left(F(\mathbf{x}^{(0)}) - F^*\right). \tag{38}$$

Notice that a sufficient condition for the convergence of a positive term series is that its partial sum series is bounded. Consequently, we deduce that $\sum_{i=0}^{\infty} \|\nabla F(\mathbf{x}^{(i)})\|_2^2$ converges.

This implies that $\|\nabla F(\mathbf{x}^{(k)})\|_2^2 \rightarrow 0$ as $k \rightarrow \infty$. Therefore, every limit point of the sequence $\{\mathbf{x}^{(k)}\}_{k>0}$ is a stationary point of the objective function $F(\cdot)$. Thus, we prove that our proposed iterative algorithm converges. ■

II. MORE VISUALIZATION RESULTS

The quantitative comparisons of SVCT reconstruction methods across various view numbers are shown in Fig. 1. It can be observed that PRIOR-S exhibits superior reconstruction performance across all test views compared to the encoder-decoder architecture RED-CNN, the dual-domain architecture CNNMAR, and the multi-scale architecture MSANet. Compared to the best-performing MSANet among benchmark methods, the average performances of PRIOR-S are improved by 1.25 dB in PSNR, 0.0110 in SSIM, and reduced by 0.0017 in RMSE, respectively. This demonstrates the effectiveness of our proposed sparse representation model-driven network that takes full advantage of the multi-scale information. Fig. 2 presents the visualization of degraded features after adding prompt features in TGM for the LACT reconstruction task using t-SNE. Specifically, Fig. 2 (a) shows that features are indistinct without using prompts. In contrast, as illustrated in Fig. 2 (b), EP enables clear and linear classification. This indicates that our explicit prompt embedding module can achieve desirable view-distinction, thereby guiding reconstructions under multiple view numbers.

To further evaluate the effectiveness of PRIOR, we supplement the comparison results of different algorithms in the SVCT reconstruction task and LACT reconstruction task, as shown in Fig. 3 and Fig. 4. The reconstruction results of various SVCT reconstruction methods from 120 views on the AAPM dataset are presented in Fig. 3. The FBP method yields images of inferior quality, rendering crucial details and structures indistinct. While RED-CNN and FBPCNN mitigate some artifacts, they lead to a loss of vital details. Moreover, CNNMAR and MSANet fall short of accurately recovering intricate details. Besides, results of DuDoTrans and FreeSeed exhibit blurred edges. Remarkably, PRIOR generates visual results with the most texture details and the least noise. Fig. 4 displays the reconstruction results of different LACT reconstruction methods from 150 views on the AAPM dataset. As we can observe, PRIOR consistently outperforms other methods. It yields enhanced organ boundary recovery, detailed structural information recovery, and boundary artifact elimination.

III. GENERALIZATION TO EXTREMELY LOW-VIEW SCENARIOS

We conduct experiments with 30 views in the multi-view SVCT reconstruction models. The primary challenge in this scenario is that severe undersampling introduces significant artifacts that struggle to mitigate. The bar charts in Fig. 5, Fig. 6 and Fig. 7 present the PSNR, SSIM, and RMSE values achieved by different multi-view SVCT reconstruction methods under the 30-views setting respectively. It can be observed that our proposed PRIOR consistently outperforms the others. Specifically, the visual comparison demonstrates that our approach achieves higher PSNR and SSIM values, reflecting its superiority in detail preservation, structural recovery, and artifact reduction. Furthermore, our method shows significantly lower RMSE, indicating a higher level of reconstruction accuracy. Moreover, PRIOR with prompt embedding outperforms the blindly trained PRIOR-B across all reconstruction metrics, thereby validating the effectiveness of incorporating explicit prompts to enhance reconstruction performance in extremely low-view scenarios. These results substantiate the robustness and generalization ability of PRIOR in extremely low-view scenarios, which can be attributed to PRIOR's unique architectural design to effectively exploit multi-scale features and view-discriminative information.

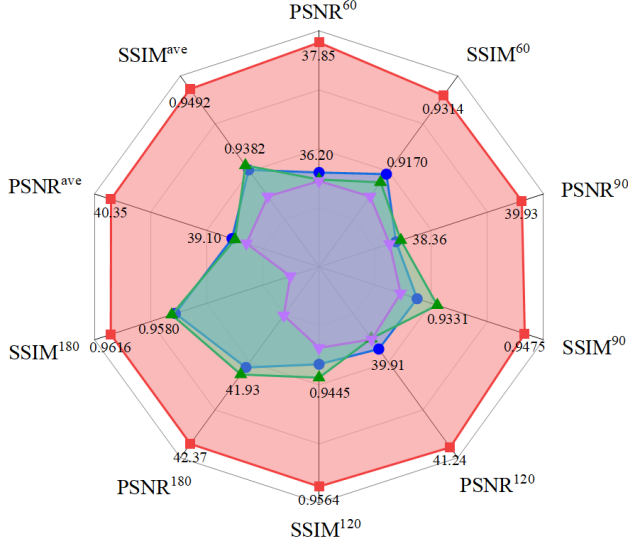


Fig. 1. Score comparisons with benchmark methods under various SVCT view numbers. We compare PRIOR-S (red) with the top 3 (blue, green and purple) benchmark SVCT reconstruction methods, i.e., RED-CNN [1], CNNMAR [2] and MSANet [3]. 60, 90, 120, and 180 are the view numbers we choose and their average scores are calculated for comparisons.

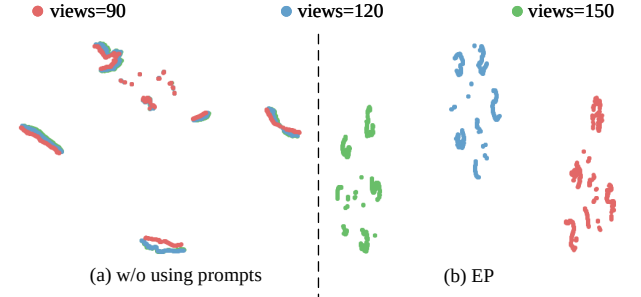


Fig. 2. The t-SNE plots of degraded features after adding prompt features in TGM for the LACT reconstruction task. Our explicit prompt strategy results in the clear distinction of features.

TABLE I

COMPUTATIONAL EFFICIENCY COMPARISONS OF FEW-VIEW CT RECONSTRUCTION METHODS ON THE AAPM DATASET IN THE MULTI-VIEW PARADIGM.

Task	Method	Inference Time (ms)	Params (M)	FLOPs (G)
SVCT (*4)	PromptIR	173.23	34.11	563.28
	AMIR	183.92	23.54	508.23
	CAPTNet	45.34	24.37	101.67
	PRIOR-B (ours)	394.13	7.06	1484.36
	PRIOR (ours)	421.83	7.18	1506.11
LACT (*3)	PromptIR	95.14	34.11	371.85
	AMIR	115.97	23.54	335.51
	CAPTNet	35.02	24.37	67.12
	PRIOR-B (ours)	349.76	9.41	1306.54
	PRIOR (ours)	375.82	9.53	1325.64

IV. INFERENCE TIME, NETWORK PARAMETERS AND FLOPS

We present a systematic evaluation of computational efficiency to assess the practical deployment potential of our proposed method. The metrics considered include inference time, network parameters, and floating point operations (FLOPs). We conduct two sets of experiments: a comprehensive comparison of few-view CT reconstruction methods on the AAPM dataset in the multi-view paradigm, and a efficiency validation of our proposed MCAF module versus standard attention.

Computational Efficiency Comparison: Table I presents the computational efficiency comparisons of few-view CT reconstruction methods on the AAPM dataset in the multi-view paradigm. While our proposed PRIOR exhibits higher inference times and FLOPs due to its relatively complex unfolding architecture, this computational investment is directly justified by their superior reconstruction quality demonstrated in previous quantitative results and visual comparisons. A key advantage of our approach lies in its exceptional parameter efficiency among multi-view models. PRIOR requires only 7.18 M parameters for SVCT and 9.53 M for LACT, far fewer than other methods like PromptIR (34.11M) and AMIR (23.54 M). This reduction in parameter count can potentially lead to lower storage costs, as fewer parameters are required to store. The minimal parameter increase between PRIOR-B and PRIOR demonstrates that our explicit prompt strategy adds negligible storage overhead while providing substantial performance gains.

MCAF Efficiency Analysis: To validate the computational performance of our MCAF module, we conduct experiments in our specific-view SVCT reconstruction model with 120 views, comparing its efficiency against a standard spatial-domain attention mechanism across varying resolutions. As shown in Table II, we observe that MCAF module consistently outperforms the

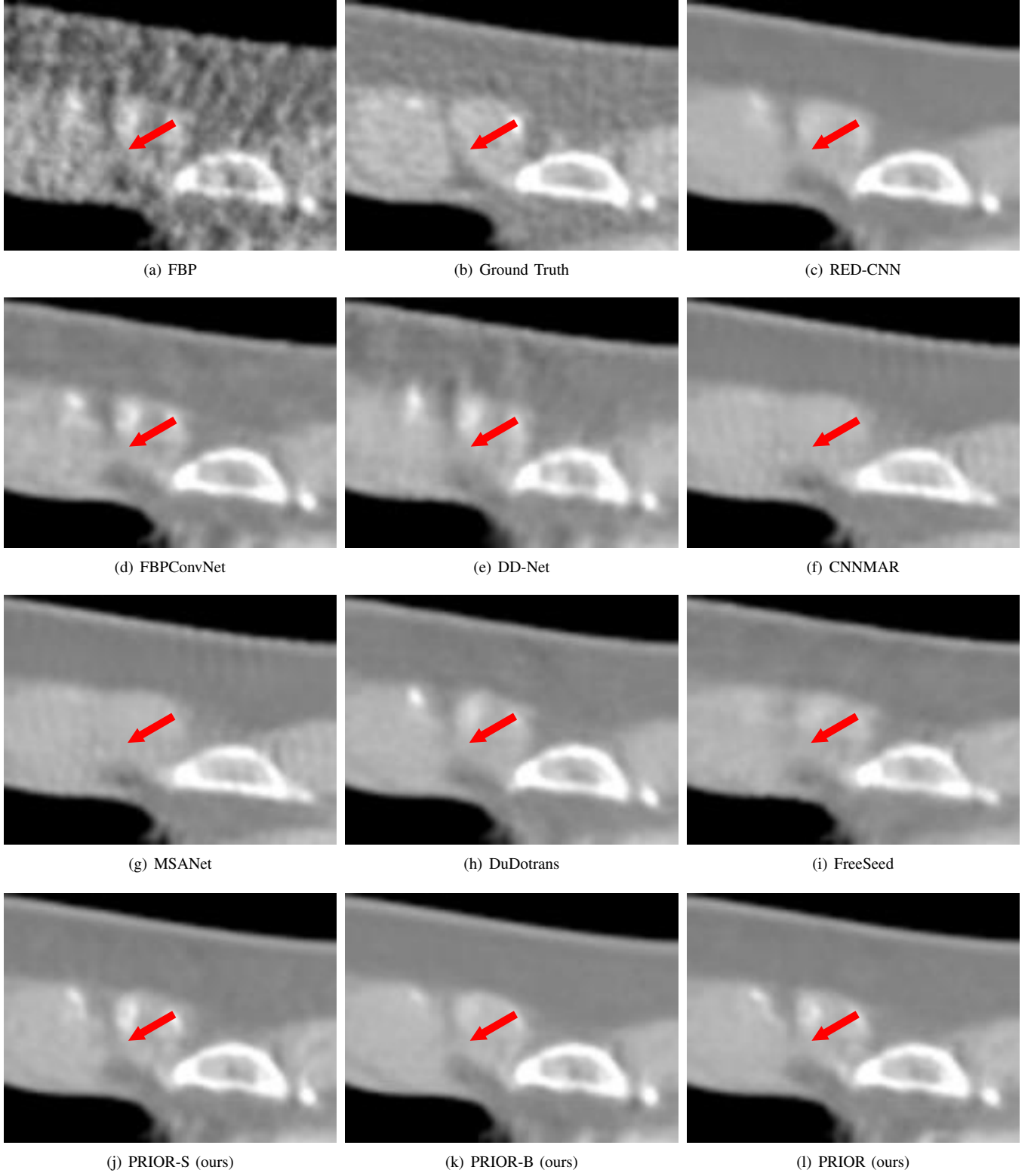


Fig. 3. SVCT reconstruction results recovered from 120 views using different methods. Regions of interest are zoomed in for better viewing. The display window of reconstructed results is $[-260, 340]$ HU for a better observation of small details.

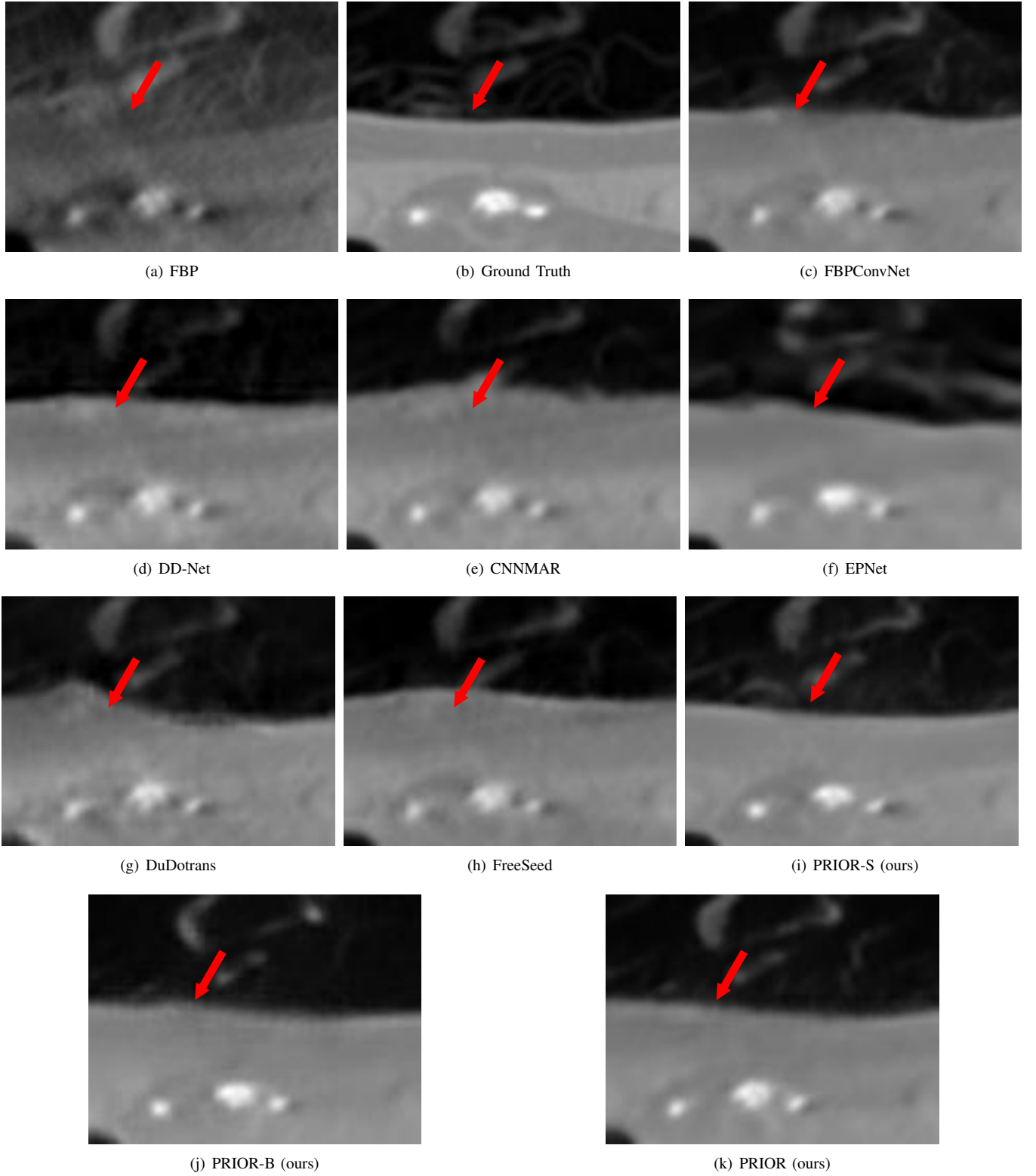


Fig. 4. LACT reconstruction results recovered from 150 views using different methods. Regions of interest are zoomed in for better viewing. The display window of reconstructed results is $[-400, 600]$ HU for a better observation of small details.

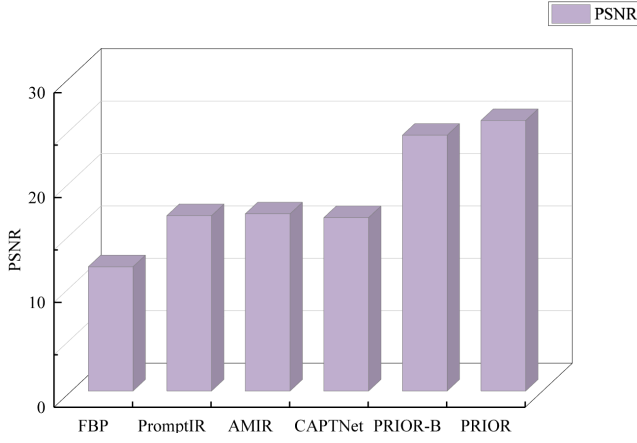


Fig. 5. PSNR (dB) $\uparrow$ of multi-view SVCT reconstruction methods recovered from 30 views.

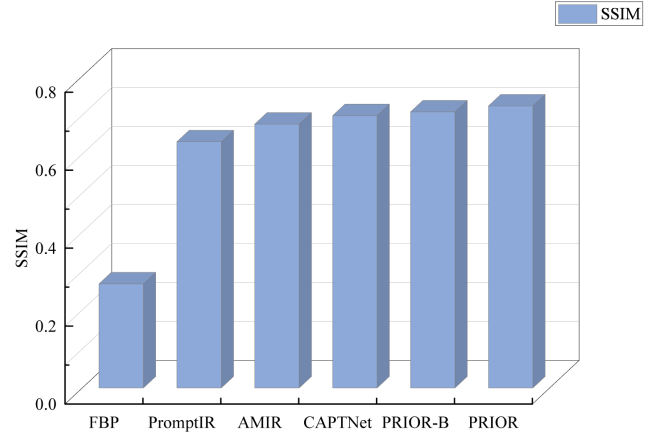


Fig. 6. SSIM $\uparrow$ of multi-view SVCT reconstruction methods recovered from 30 views.

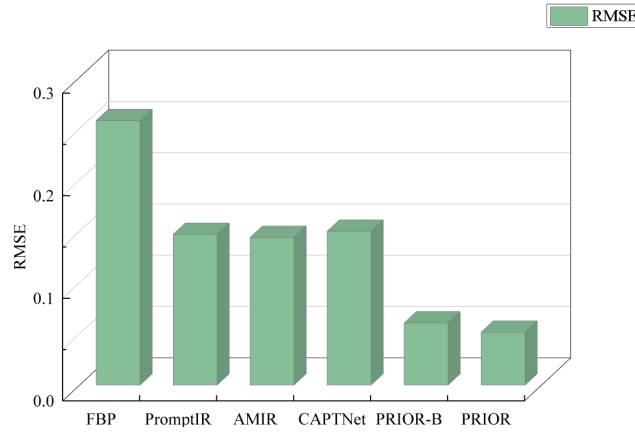


Fig. 7. RMSE $\downarrow$ of multi-view SVCT reconstruction methods recovered from 30 views.

TABLE II
COMPUTATIONAL EFFICIENCY COMPARISONS OF MCAF MODULE VERSUS STANDARD ATTENTION ACROSS DIFFERENT INPUT RESOLUTIONS.

Input Resolution	Module	Inference Time (ms)	Params (M)	FLOPs (G)
256×256	Standard	266.31	6.90	371.39
	MCAF	218.91	6.86	371.09
416×416	Standard	398.91	6.90	981.71
	MCAF	308.04	6.86	979.91
512×512	Standard	484.43	6.90	1505.57
	MCAF	352.59	6.86	1484.36

standard attention module in terms of computational efficiency across all tested resolutions. At the 256×256 resolution, MCAF achieves a 17.8% reduction in inference time while maintaining comparable parameters and FLOPs. This efficiency gain scales with the resolution, resulting in a speedup of 22.8% at 416×416 resolution that further increases to 27.2% at 512×512 resolution.

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